

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2025****Course:** Mechanical Engineering **Branch:** Mechanical Engineering**Semester:** VI**Subject Code & Name:** BTMEC605C_Y19 Renewable Energy Sources**Max Marks:** 60**Date:**10/07/2025**Duration:** 3 Hr.**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is considered a non-renewable energy resource? a. Solar energy b. Wind energy c. Coal d. Biomass	Knowledge	1
2	Which energy resource has the largest proven reserves in India? a. Coal b. Natural Gas c. Uranium d. Crude Oil	Knowledge	1
3	Nuclear fission involves: a. Splitting of a heavy nucleus into lighter nuclei b. Combining of light nuclei into a heavier nucleus c. Conversion of solar energy into electricity d. None of the above	Knowledge	1
4	Which instrument is commonly used to measure solar radiation? a. Pyranometer b. Anemometer c. Hygrometer d. Thermometer	Knowledge	1
5	Which material is commonly used for the absorber plate in flat plate solar collectors? a. Aluminum b. Copper c. Plastic d. Stainless steel	Understanding	1
6	Which of the following is an advantage of solar energy? a. Emits greenhouse gases b. Limited availability during day c. Renewable and non-polluting d. High operational cost	Understanding	1
7	What is the role of tracking systems in concentrating collectors? a. To reduce heat loss during operation b. To maintain alignment with the sun's position c. To circulate the working fluid d. To measure solar radiation	Understanding	1
8	What is the most common application of solar water heating systems?	Application	1

	a. Industrial steam generation	b. Domestic hot water supply	c. Air conditioning	d. Electricity generation		
9	Flat plate collectors are typically tested for:				Application	1
	a. Weight and durability	b. Electrical conductivity	c. Thermal performance and efficiency	d. Noise level		
10	What is the primary source of wind energy?				Application	1
	Ocean currents	Rotation of the Earth	Uneven heating of the Earth's surface by the sun	Gravitational pull of the moon		
11	What is the primary purpose of a bio-conversion plant?				Application	1
	a. Generate solar power	b. Convert biomass into usable energy	c. Store wind energy	d. Extract fossil fuels		
12	What is the source of geothermal energy?				Analysis	1
	a. Heat from the sun	b. Heat generated by the Earth's core	c. Combustion of fossil fuels	d. Chemical reactions in the ocean		
Q. 2	Solve the following.					12
A)	Enlist various types of energy resources available and recommend future energy resource.				Application	6
B)	What are energy resources? Discuss the current status of energy conversion in India.				Understand	6
Q.3	Solve the following.					12
A)	What are the different types of concentrators used in concentrating solar collectors? Describe each briefly.				Remember	6
B)	What are the common methods used for measuring solar radiation?				Understand	6
Q. 4	Solve Any Two of the following.					12
A)	Explain flat plate solar collectors and describe the selection criteria for choosing suitable flat plate collectors.				Understand	6
B)	How are flat plate solar collectors tested to evaluate their performance and efficiency?				Understand	6
C)	What are the basic components of a solar water heating system, and what is the working principle behind it?				Apply	6
Q.5	Solve Any Two of the following.					12
A)	State the key advantages of wind energy as a renewable energy source.				Apply	6
B)	Define biomass and explain its various forms with suitable examples.				Understand	6

c)	List and explain the different types of windmills in detail.	Understand	6
Q. 6	Solve Any Two of the following.		12
A)	What is tidal energy? Explain the process of its generation.	Understand	6
B)	What is geothermal energy? Describe how it is harnessed for electricity production.	Apply	6
c)	What is hydroelectric energy? Explain the method of its generation.	Understand	6
	*** End ***		